Hate Speech Detection using Machine and Deep Learning

Social media is very popular for communicating and sharing ideas. But many bad things are also shared, like hate speech. Hate speech is language that attacks people because of their group, such as their race, religion etc. This kind of speech makes online spaces unsafe and can cause real-world problems. It is very hard to automatically find and stop hate speech because the hate language is always changing, and context is important. Researchers are working hard to build algorithms that can detect this language fast and efficiently.

One of the simplest ways to detect hate speech is using simple ML models. The authors in the paper[1] compared different ways to extract features with standard machine learning models. They used TF-IDF and Bag of Words (BoW) as feature extraction techniques. The machine learning models were Logistic Regression, Naïve Bayes, Decision Tree, and KNN. They detected hate speech from live tweets on Twitter. The Decision Tree model had the best accuracy (92.43%) with TF-IDF. This was 2% better than Logistic Regression. This shows that simple feature methods can work reasonably well.

But, this can be improved more by using transfer learning. This method lets a model use knowledge it learned from one task to help with a new one. The authors in [2] worked on transfer learning for hate speech detection. They did not use the usual method of transfer learning. Normally, weights from one model are moved to a new model. Instead, they fine-tuned the embedding weights from ELMo and GLoVe by solving both tasks at the same time. This created one space for representation. They also made a tool called the Map of Hate. This tool is a visualization that helps compare datasets. It can find labeling errors, which helps human moderators. The accuracy gain was small (1% to 2%). They explained that it was because the two datasets had different labeling rules.

But improving the result does not hold very long as people get smart and start using misspellings and coded words for hate speech. Researchers in paper [3] wanted to solve this exact problem. They created unsupervised domain-specific

word embeddings to use as features. They used a Bi-LSTM as the classifier model. They collected a large dataset, including one with known hate speech words. They compared their model against a pre-trained BERT model. They also discussed how the BERT model makes its predictions using LIME. This helps explain why the model made a certain prediction.

The challenges to detect hate speech does not end there. Hate speech differs from language to language. This paper [4] focused on detecting hate speech in the Bengali language. They collected a dataset of 7,425 Bengali comments from Facebook pages. They used TF-IDF vectorization and word embeddings for features. They made a module to find emotions from emojis and emoticons in Bengali. They used a CNN as an encoder and then used LSTM, GRU, and attention mechanism as a decoder. The attention mechanism helps the model focus on key words. Their model showed better performance than three other papers.

Even after detecting hate speech from different languages the challenge does not end. The authors in the paper[5] created a new dataset called BANTH which is the first multilabel hate speech dataset for transliterated Bengali language. They collected 37,350 comments from YouTube. They pre-trained encoder-based models on transliterated Bengali. The models achieved very high accuracy. They tested methods like zero-shot, few-shot, and introduced a new method called Translation Prompting using different GPT models. Making new datasets like this helps researchers to find good solutions for the new challenges.

To detect hate speech with high accuracy, models in general get complicated. When models get complicated, it is important to understand why they make certain choices. This is called explainability. Researchers in study[6] focused on Explainable Artificial Intelligence (XAI). They wanted to understand how different machine learning models made decisions for hate speech detection. They tested hybrid models (BERT+MLP and BERT+ANN) and compared them with simpler machine learning models. They found that the hybrid models worked better. They also tested these models on the Google Jigsaw dataset. Understanding the reasons behind the model's prediction is important for trust.

References

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